Started on	Wednesday, 27 November 2024, 2:17 PM
State	Finished
Completed on	Wednesday, 27 November 2024, 2:47 PM
Time taken	30 mins 1 sec
Marks	1.33/3.00
Grade	**1.33** out of 3.00 (**44.46**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Not answered

Marked out of 1.00

Bob is using the RSA encryption scheme with public key equal to $(n, x) = (3763, 2029)$.

a) Bob's RSA key is weak. Break his encryption and find his Bob's private key:

$y =$ 98

Your last answer was interpreted as follows:

$$98$$

It might help to know that the following is a list of the first prime numbers:

$$[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79]$$

b) Alice uses Bob's public key to encrypt a message, which she then sends over a public channel.

The encrypted message is

$$M = 2873$$

Find the unencrypted (plain text) message

$m =$

The following calculator can be used for computing exponents $\mod (n)$:

Base:

Exponent:

Modulus:

Result:

Question **2**

Correct

Mark 1.00 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

$$[000100, 001010, 011010, 001000]$$

.

$l_{min} =$ 1

Your last answer was interpreted as follows:

$$1$$

$l_{max} =$ 4

Your last answer was interpreted as follows:

$$4$$

Question **3**

Partially correct

Mark 0.33 out of 1.00

We use the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

to define the encoding function

$$E : (\mathbb{Z}/2)^{\times 4} \to (\mathbb{Z}/2)^{\times 8}$$

by the formula

$$E(w) = w \cdot G\,.$$

An encoded message $c = E(w)$ is transmitted over a noisy channel and received as

$$r = 10110101\,.$$

a) Compute the syndrome of $r =$ 0010

Your last answer was interpreted as follows:

$$0010$$

b) Compute the error pattern e, so that $r = c + e$.

$e =$ 00000110

Your last answer was interpreted as follows:

$$00000110$$

c) Compute the encoded message $c =$ 10110011

Your last answer was interpreted as follows:

$$10110011$$

Syntax guide: Give your answers as bitstrings E.g. if you think the answer to a question is "1011", then you should enter **1011** .

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 112

Signing code: 46006

Your last answer was interpreted as follows:

$$46006$$

◄ Test 3 (topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod(n) and Lagrange's theorem)

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